code via an ancilla. The LFR R_{unload} of that is given by:

$$\begin{aligned}
R_{unload} = 2\Bigg(& \frac{\sqrt{n_m}}{5} R_S \\
& + \sqrt{n_a} R_L(n_a) \\
& + \frac{\sqrt{n_m}}{5}\left(R_L(n_a) + \frac{n_m}{25} R_L(n_m)\right) \\
& + \frac{\sqrt{n_m}}{5}\left(R_L(n_a) + R_S\right)\Bigg)
\end{aligned} \tag{14}$$

Now for Bell-Measurement, we require $4\frac{\sqrt{n_m}}{5}$ QEC cycles [42]. The LFR R_{BM} for this is given by:

$$R_{BM} = 4\frac{\sqrt{n_m}}{5} R_S \tag{15}$$

This generates a correlation between Alice and Bob.

Total LFR: So, therefore, the LFR R_{tot} of the overall final correlation between end users Alice and Bob is given by:

$$R_{tot} = R_{BM} + R_{unload} + R_{store} + 2R_{net} \tag{16}$$

And the success probability F_{final} of this correlation is given by:

$$F_{final} = 1 - R_{tot} \tag{17}$$

Logistics: Logical Bit-rate r_L per qLDPC block is given by:

$$r_L = \frac{n_m}{25 T_4} \tag{18}$$

This is the number of logical qubits from the qLDPC block unloaded and measured per unit time. It takes T_4 amount of time to unload one entire qLDPC patch consisting of $\frac{n_m}{25}$ logical qubits.

Let r_e be the desired target entangled bit (E-Bit) rate. Then, the number N of parallel qLDPC blocks required to achieve r_e is given by $N = r_e/r_L$. The blocks get depleted every T_4 units of time. Therefore, we need to replace N blocks every T_4 units of time. Let's say a truck or a train car has the carrying capacity of B blocks, where $B = n_t/n_m$ where n_t is the total number of physical qubits the truck or train car can carry. We need N/B trucks to be released by Charlie Inc. every T_4 units of time, for one qATM destination. if we are assuming S number of qATM destinations, then the number of trucks n_{truck} released by Charlie Inc. every T_4 units of time is given by $n_{truck} = SN/B$.

The life-cycle of a truck involves T_2 units of time to load the qLDPC patches, T_3 time to transit, T_4 time to unload and get the logical qubits measured, and T_3 again to move back to Charlie Inc.'s hub for the next round. So, the total number of trucks $N_{truck_{tot}}$ required for the whole network to run continuously is given by:

$$N_{truck_{tot}} = \frac{SN}{(n_t/n_m)} \frac{T_2 + 2T_3 + T_4}{T_4} \tag{19}$$

V. RESULTS AND DISCUSSION

The required E-bit rate (r_e), or bandwidth, is a critical parameter in our system design. Contemporary communication standards provide context for our bandwidth requirements. For instance, voice communications necessitate 87 kbps, standard definition video conferencing requires 128 kbps, and high-definition video conferencing demands up to 4 Mbps [43]. In the realm of text messaging, an average SMS of 120-160 characters consumes approximately 16,000 bits [44], which we consider for our analysis. Note that this is somewhat contrived, but it gives us an anchor to do our analysis. To determine r_e, we consider several factors: the frequency of one-time pad recharging, the practical time a user might wait at a quantum ATM while recharging (t_{wait}), and the average number of bits needed between refills (b_{req}). The number of end-users at a destination (n_{users}) is constrained by the number of available time slots in a day, calculated as 86,400 seconds divided by t_{wait}. Consequently, we express r_e as b_{req}/t_{wait}.

We estimate r_e for SMS usage as an illustrative example. Assuming an average person sends 85 texts daily [45], each requiring 16,000 bits, we calculate b_{req} as $85 \times 16,000 = 1,360,000$ bits. With a t_{wait} of 10 minutes (600 seconds) every day, r_e equals 1,360,000/600 = 2.3 kbps or 2.3 KHz.

The gate error rate p_g for neutral atom architectures varies from 0.001 to 0.005. Quantinuum, a quantum computing company, has demonstrated a one-qubit gate fidelity of 0.999979 and a two-qubit gate fidelity of 0.99914 [46]. Therefore it would be reasonable to consider $p_g = 1 - 0.9992 = 0.0008$ as a near-term realizable optimistic estimate.

Idling Errors: Idling errors can be modeled by adjusting the two-qubit gate error rate p_g as follows [27]:

$$p_g \to p_g \left(1 + \frac{3t_r(n)}{0.005 T_c}\right) \tag{20}$$

where $T_c = 10$s is the coherence time of neutral atom qubits. If we plug in a worst-case scenario of a patch size of $100,000$, we see that p_g gets scaled up by a factor of 1.2754 i.e. by 27.54%. Considering our original estimate of $p_g = 0.0008$, the new p_g becomes $p_g = 0.0008 * 1.2754 = 0.001$, which we use for our analysis.

Unlike most other architectures, neutral atoms have significantly slower 1-qubit gates than 2-qubit gates. Hence we take the 1-qubit gate time $t_g = 2\mu s$ [40]. Considering the form factor of Pasqal's processor [47] and their claim of producing a 10^4 qubit unit by 2026, we take a rough estimate for our calculations that any truck or train car in the near future can carry a total of $n_t = 1,000,000$ physical qubits. We also set the number of qATM destinations $S = 5$. In this article, we assume intra-metropolitan distances, with typical transport times ranging from one to three hours. We take the teleportation ancilla HGP patch size $n_a = \frac{n_m}{25}$ because it maintains the same distance as the qLDPC code and

QEC Code	Number of Trucks	Cost Per Bit
qLDPC	7444	USD $1.40
Surface Code	31920	USD $5.99

Table I. Table showing the number of trucks required and the cost of each bit sold rounded to the nearest cent, for qLDPC-based protocol and a pure Surface-Code-based protocol. Here, we have taken the transit time T_3 to be 90 minutes, network bandwidth to be 2.3 kHz, hourly rent per truck as USD $150, and the total number of end-nodes to be 5. The cost is calculated as per Eq. 21.

the surface code, and its edge is as wide as the size of the logical operators in the qLDPC code [27].

From Figure 3, we can see that to achieve a lower total logical error rate R_{tot}, we need larger qLDPC patches. Larger patches have greater code-distance, hence higher tolerance to errors. To achieve greater than 80% fidelity (i.e. $R_{tot} < 0.2$), we need patch sizes of at least about 35,000 physical qubits (1400 logical qubits). Patch sizes over 53,000 physical qubits (2120 logical qubits) can achieve fidelities of over 90% while 80,000-qubit-sized patches (3200 logical qubits) can give greater than 95% fidelity. Higher the fidelity, lower is the number of bits wasted for standard reconciliation and privacy amplification, and with fidelities over 95%, this number becomes negligible [48]. Pasqal, an enterprise dealing with neutral atom quantum architectures, claims that devices with a capacity of 10,000 physical qubits will be a reality in the next couple of years [47], so it would be reasonable to assume larger patch sizes of the order of high 10^4 physical qubits to be achievable in the next few years.

From Figure 4, we can observe that the protocol can tolerate longer transport times T_3 with larger patch sizes. This is because larger patch sizes have greater logical error tolerance, and hence can be transported to longer distances. It can also be observed that the plot truncates below a certain patch size when the value of T_3 comes down to zero. This is because, for any n_m below that patch size, all terms in R_{tot} except the R_3 term push the R_{tot} greater than the target error rate. Due to this, the solver function in our plot code plugs in a negative T_3 to pull down the R_{tot} to match the target. This means that the target R_{tot} is not achievable for patch sizes below this cutoff size. From both protocols, we see that the error rate R_{tot} increases with an increase in transport time due to a greater number of error correction cycles.

From Figure 5, we can see that for a given transport time (90 minutes here), the number of trucks or train cars required reduces with an increase in the tolerable error rate R_{tot} of the network. This is because the greater the tolerable error rate R_{tot}, the smaller the patch size. Smaller patch sizes are of lower code distances and hence require fewer rounds of error correction. This translates to faster unloading times, requiring a lesser number of patches in parallel, and ultimately requiring a lower number of trucks. For fidelities below 95% (i.e. $R_{tot} > 0.05$), we can run this network with under 10,000 trucks for 2300 Hz bandwidth. This is comparable to the total number of buses or train cars in an average city public transport.

Figure 6 compares the qLDPC network with the corresponding surface-code network, in terms of number of trucks, and additionally, table I also lists the operational unit economics C_o in terms of the cost per bit, which is a combination of transportation cost C_t and the cost C_q of maintaining the quantum infrastructure i.e.:

$$C_o = C_t + C_q \tag{21}$$

The transport cost C_t calculated as follows:

$$C_t = \frac{R_h N_{truck_{tot}}}{3600 r_e S} \tag{22}$$

where R_h is the hourly rent of one truck, which we assume to be USD $150, $N_{truck_{tot}}$ is as described in Eq. 19, r_e is the E-bit rate or the bandwidth of the network, and S is the number of network nodes or end-stations. Next, the total cost C_q of maintaining the quantum infrastructure is given by:

$$C_q = \frac{2\frac{n_t}{n_m} N_{truck_{tot}} C_m}{r_e S \times 86400 \times 365} \tag{23}$$

where C_m is the yearly cost of maintaining a single qLDPC memory patch containing n_m physical qubits. The numerator represents the yearly cost of maintaining all the qLDPC memory patches in the network, where the factor 2 arises from the fact that we are dealing with Bell-pairs, and n_t/n_m is the number of quantum computer units on a truck. The denominator represents the number of bits emenated per year by the network. We consider an estimate of $C_m = \$2,000,000$ per year per quantum device unit [49]. For surface codes, the quantity n_m would represent the number of physical qubits in a single quantum device, which would contain multiple surface code patches. We keep n_m same for both qLDPC and surface codes.

The number of transport vehicles required by the qLDPC network is one order of magnitude lower than the number of vehicles required by the surface code protocol, and so is the cost. For example, for a minimum fidelity of 0.92, patch size $n_m = 60,000$, and a transit time of 90 minutes, the qLDPC network requires 7444 vehicles to run and costs C_o =USD $1.40 per bit, whereas an equivalent surface code requires 31920 vehicles and costs C_o =USD $5.99 per bit, about 4 times more. This is due to the constant encoding rates of qLDPC codes regardless of the tolerable logical error rate, unlike surface codes whose encoding rate worsens with the required tolerable logical error rate.

Nevertheless, even with qLDPC codes, the cost of $1.40 per qubit would mean an average of $22,400 per text message at 16,000 bits per message, which is extremely expensive, thus making the protocol in its current form commercially infeasible. High number of trucks

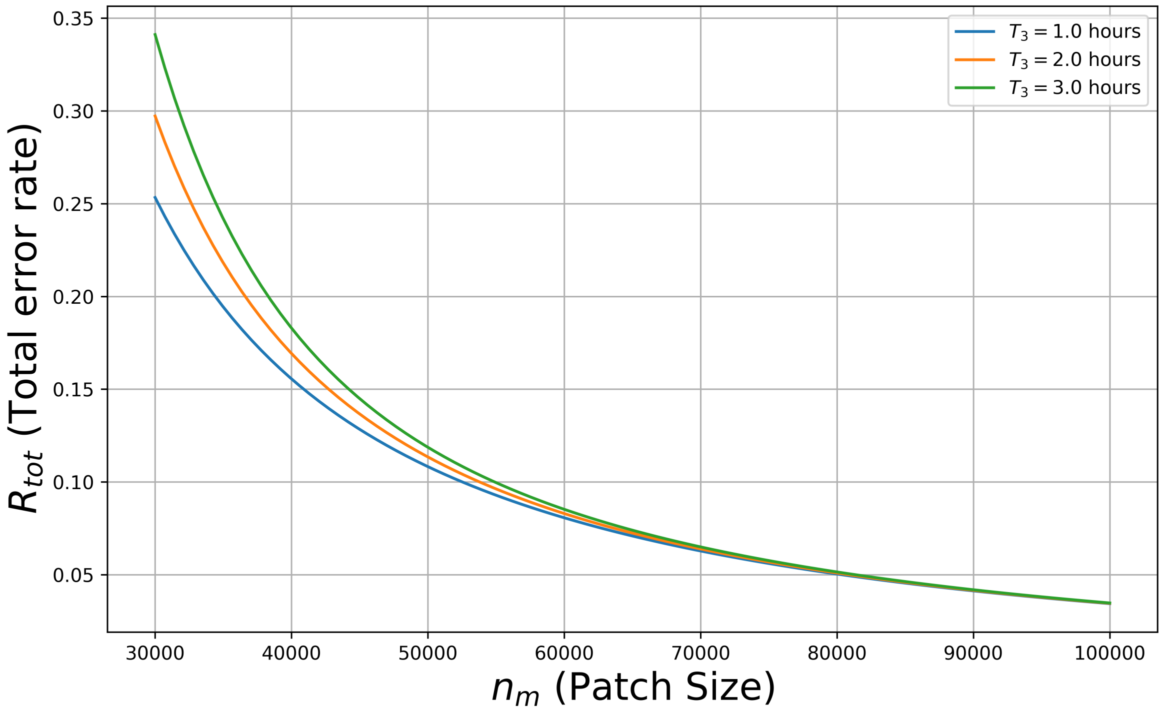

Figure 3. The overall Logical Error Rate R_{tot} plotted against qLDPC patch size n_m for various transport times T_3, showing an inverse relationship between the total logical error rate R_{tot} and the size of qLDPC patches. To obtain fidelity levels exceeding 80% (corresponding to $R_{tot} < 0.2$), patch sizes of approximately 35,000 qubits or more are necessary. Further improvements in fidelity can be achieved with larger patches: those containing over 53,000 qubits can reach fidelities surpassing 90%, while patches comprising 80,000 qubits or more are capable of attaining fidelity levels above 95%.

and consequently high costs per qubit are both mainly attributable to the fact that neutral atom architectures have their physical qubits sparsely spaced in the lattice, and their gate execution times are relatively slow. If we can implement qLDPC codes on faster, more condensed architectures in the future, such as the silicon quantum dot qubits for example [50], we can potentially reduce the number of trucks and consequently the cost per bit by about 3 orders of magnitude.

VI. CONCLUSION

Our analysis shows that in the near term, it is possible to build a multi-node Delayed-choice network, all separated by intra-metropolitan distances (with transport times of one to three hours), with a central Charlie Inc. who uses $O(10^3)$ vehicles to transport and sell entanglement-based correlations to end-users with classical hardware (such as a smartphone) at the end nodes. Assuming that Charlie Inc. has the requisite quantum memory and storage, and it is possible to distribute entanglement at a rate faster than the entanglement consumption, then many end-users can top up their results on classical storage devices, and use them at their convenience, topping up as required. This commodifies entanglement, paving the way to large-scale commercial quantum networks catering to users with non-quantum hardware. The qLDPC codes, with their compact constant encoding rates, play a major role in establishing this feasibility.

In our analysis, we adopt certain conventions and assumptions based on recent research in quantum error correction codes. The relationship between the number of physical qubits in Low-Density Parity-Check (qLDPC) memory and Surface Code is a key consideration. We assume that a qLDPC memory with n_m physical qubits is connected to a surface code with $n_m/25$ physical qubits for teleportation. This ratio is consistent with the simulations presented in ref. [27], although it is not explicitly stated as a requirement. While it is theoretically possible to use surface codes of various sizes, we adhere to this ratio for consistency with existing literature.

Regarding the teleportation scheme for transferring logical qubits between qLDPC and surface codes, our analysis focuses on Hypergraph Product (HGP) codes. Ref[27] specifically defines this teleportation scheme for HGP codes, but not for Lifted Product (LP) codes.